MATRIX-GRID · RENDERING JANK

# The grid that moved after it painted.

`matrix-grid` drew its columns from its own cell text. When the web font landed a frame after first paint, every column re-solved and every cell slid sideways — with nothing overflowing for a fit gate to catch.

WHAT WAS MOVING

# Column widths were a function of the font that had not loaded yet.

**Measured, not inferred**

Same deck, same Studio, fonts blocked against fonts allowed. Blocked, the columns never moved.

**The five columns moved**

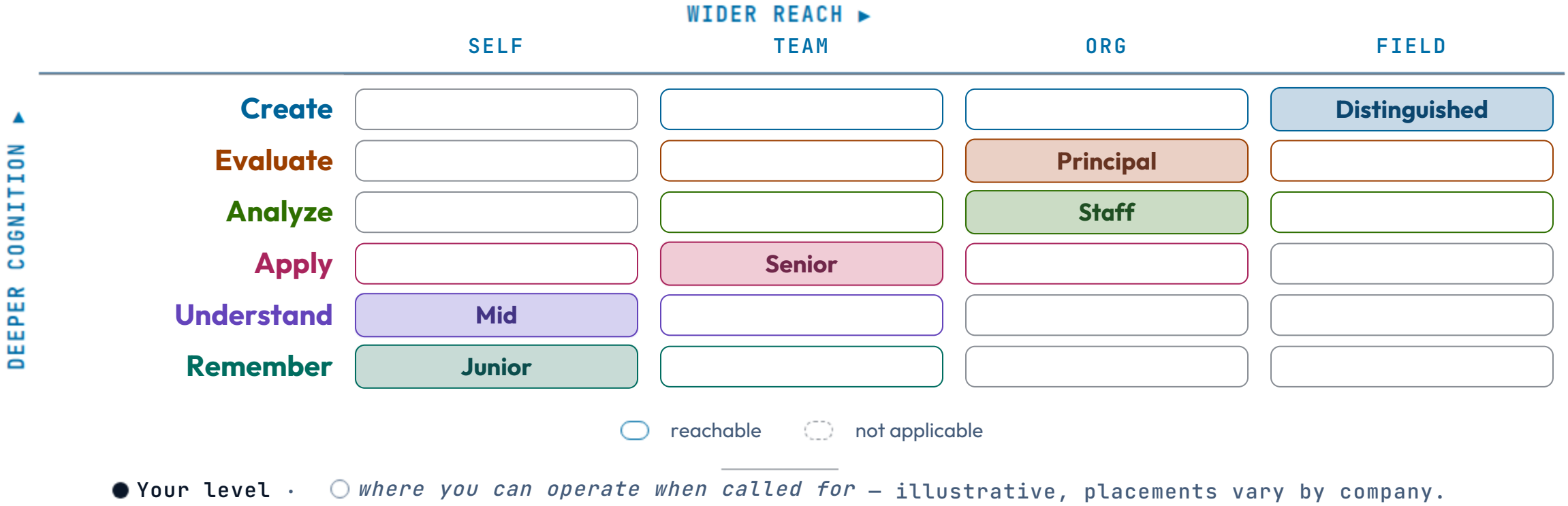
`258.2 · 165.7 · 175.5 · 216.4 · 304.2` on first paint. `248.7 · 176.9 · 178.9 · 218.9 · 296.6` one frame later.

**Invisible to every gate**

The table was 1120px wide before and after. Nothing overflowed, so nothing was reported.

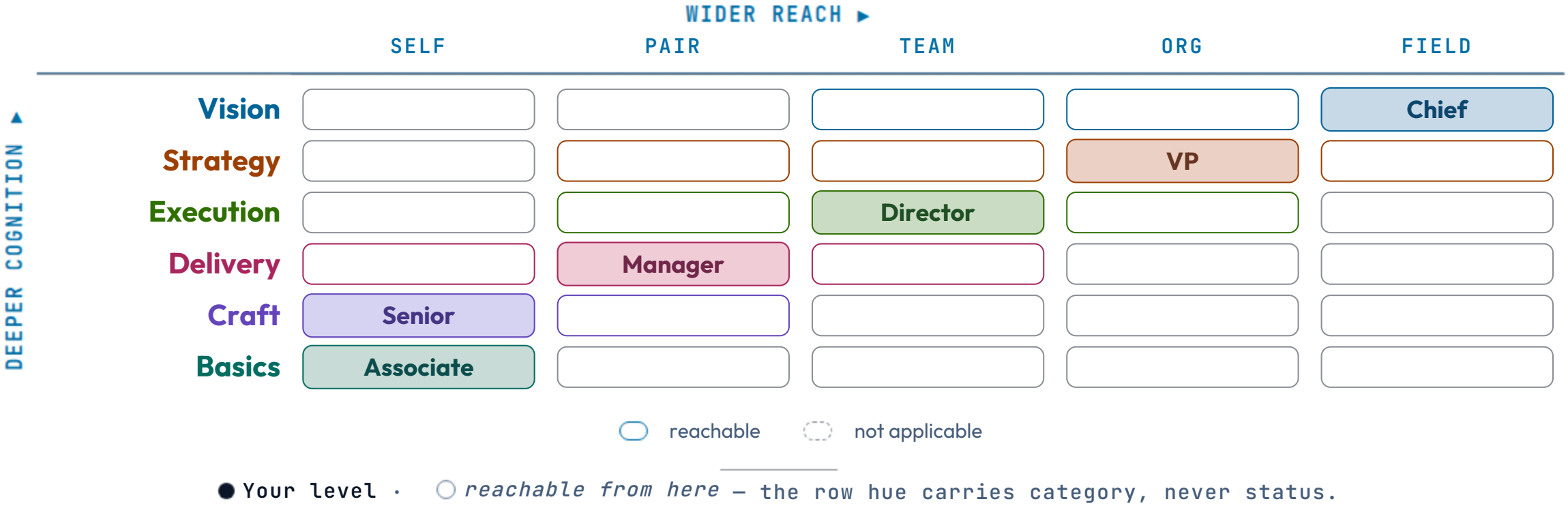
# Your level is a cell, not a rung.

*Every column is the same width, so the reach axis reads as equal steps.*



# Six categories by five columns is the grid's practical ceiling.

*The widest label no longer buys width from its neighbors.*



HOW IT IS HELD STILL

# `table-layout: fixed` takes the text out of the geometry.

**Geometry stops reading text**

A fixed table divides its width by declaration. A late font changes the glyphs, never the column.

**The sibling already did it**

`roadmap`, the other table in this bucket, has been fixed-layout all along, for this reason.

**Equal steps read truer**

Under `auto` the reach axis was sized by the longest label: 165.7px against 304.2px, an 1.8x spread.

ONLY WIDE IS WIDE ENOUGH

# Every narrower family still has to negotiate.

### Measured as text advance

`scrollWidth` on a centered flex box misses the half that overflows left. A Range over the text node is the honest unit — and it reads `-25.2px` where `scrollWidth` reported a 2px rounding artifact.

### Square fails like portrait, later

`--canvas-scale` raises the type as the box narrows. `Distinguished` clears its cell by `+75.9px` at wide, misses by `-25.2px` at square and `-44.5px` under `mode: sketch`.

ONLY WIDE IS WIDE ENOUGH

# And the type has nowhere left to go.

**The type floor is reached**

`--fs-meta` is the floor of the twelve-token scale, so the step the narrow arm already takes is the last one available.

**So only wide takes** fixed

Square, tall and strip keep `auto` and render byte-identically. The shift stays live there, and `scale-2xl` leaves wide only `+19.7px`.

RELATED

# See also.

`roadmap` — the sibling table, fixed-layout since it shipped

`obligation-matrix` — rows × columns of pass/partial/exempt status

`engineering/jank.md` — does the layout stay put as content grows